Computer-Based Data Processing

Lab 1 – Getting Started with LibreOffice Calc



Objectives

The aim of this first lab is to explore the main operations used to manipulate data in a spreadsheet.

Throughout the lab, we will use a dataset containing several hundred **space missions carried out between 1957 and 2022**.

By the end of this lab, you should be able to:

- read and explore a data table;
- write and fill down a formula;
- calculate simple statistics;
- use a condition with `IF()`;
- combine several conditions;
- use an absolute reference.

Start by downloading the `space_missions.csv` file from the course platform, then open it with LibreOffice Calc.

Important: check that LibreOffice Calc uses English locale settings. Go to `Tools > Options > Languages and Locales > General`, then select `English (USA)` under `Locale setting`.

1 Exploring the dataset

Each row in the table corresponds to a space mission. The columns describe the organisation responsible for the launch, the location, the date, the rocket used, its price, and the mission outcome.

Take a few minutes to explore the table. In particular, identify the column containing the mission outcome and examine the different values it can take.

Some information is missing for certain missions. We will return to this issue in a later lab.

Your first formula

Add a new column to the right of the table called `Successful mission`.

In the first data row, write a formula that displays:

- `Yes` if the mission outcome is `Success`;
- `No` otherwise.

The `IF()` function follows this structure:

```
=IF(condition; value_if_true; value_if_false)
```

For example:

```
=IF(A2>10; "Yes"; "No")
```

Once your formula works correctly, fill it down through all the rows of the table.

Tip: instead of dragging the formula down to the last row, double-click the fill handle at the bottom-right corner of the cell.

2 Describing the data

We will now calculate some simple information about the dataset. Create a small summary area to the right of the table.

2.1 Counting missions

The `COUNTA()` function counts the number of non-empty cells.

Use it on a column that contains a value for every mission to determine the number of missions in the dataset.

A range such as `A2:A500` can also be replaced with `A:A` to refer to the entire column.

2.2 Examining rocket prices

The `Price` column gives the rocket price, expressed in **millions of US dollars (M$)**.

Using the functions:

- `MIN()`
- `MAX()`
- `AVERAGE()`

determine:

1. the minimum price;
2. the maximum price;
3. the average price.

Is a price available for every mission? Observe how Calc handles empty cells.

2.3 Going a little further

The `SMALL()` and `LARGE()` functions retrieve a value according to its rank.

For example:

```
=SMALL(A:A; 2)
```

returns the **second-smallest value** in column A.

Similarly:

```
=LARGE(A:A; 2)
```

returns the **second-largest value**.

Use these functions to find:

1. the second-lowest price;
2. the second-highest price.

3 Making decisions with conditions

We will now enrich the dataset by creating new information.

3.1 Success or failure

You have already created a column indicating whether a mission was successful.

Now modify this formula to distinguish three situations:

- `Success` if the outcome is `Success`;
- `Partial failure` if the outcome is `Partial Failure`;
- `Failure` in all other cases.

To do this, you can nest several `IF()` functions.

3.2 Combining several conditions

Add a new column called `Successful mission with active rocket`.

It should display `Yes` only if:

- the mission was successful;
- **and** the rocket is still active.

The `AND()` function checks several conditions simultaneously:

```
=IF(AND(A1>10; B1<5); "Yes"; "No")
```

Then fill your formula down through the entire dataset.

4 Normalising a variable

We now want to transform the prices so that they fall between 0 and 1.

For a value x, we use:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}.$$

4.1 Preparation

In two separate cells in your summary area:

1. calculate the minimum price;
2. calculate the maximum price.

Then add a new column called `Normalised price`.

4.2 First attempt

In the first row with an available price, apply the normalisation formula.

Then fill the formula down.

Look carefully at the references used in the following rows.

What happened?

4.3 Absolute references

When you copy a formula, a standard reference such as `A1` changes automatically. This is called a **relative reference**.

To prevent a reference from changing, use an **absolute reference**, for example:

```
$A$1
```

Correct your normalisation formula so that the references to the cells containing the minimum and maximum remain fixed, then fill it down.

Check that:

- the lowest price gives a value close to 0;
- the highest price gives a value close to 1.

Further practice

If you have finished the lab, add a column that classifies each mission according to its era:

- `Early space age` before 1980;
- `Modern era` from 1980 to 2000;
- `Contemporary era` after 2000.

Use several nested `IF()` functions.

Review

During this lab, you worked with a real dataset in LibreOffice Calc.
You learned how to:

- write and fill down formulas;
- use `COUNTA()`, `MIN()`, `MAX()`, `AVERAGE()`, `SMALL()`, and `LARGE()`;
- create new information with `IF()`;
- combine several criteria with `AND()`;
- use absolute references;
- apply a mathematical formula to an entire column.